



Credit Analysis from First Principles

A Private Credit Handbook — Ratios, Red Flags, Repayment Analysis, and the A-to-Z of Every Strategy

Direct Lending · Fund Finance (Subscription / RCF) · NAV Financing · Trade Finance · Venture Debt

Companion workbook: **Credit_Analysis_Worked_Example.xlsx** (Meridian Foods case study)

Prepared July 2026 · Internal training material · All examples fictional

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HOW TO USE THIS PACK

Read Parts 1–5 once, slowly — they are the foundation and they apply to every strategy. Then read the strategy chapters (Part 6) that match your book. Then open the Excel workbook and work the Meridian case yourself *before* reading Part 8, which gives the model answer. Parts 9–10 are reference material for the day job. The single most important idea in the entire pack appears in Part 1 and is repeated deliberately throughout: **debt is repaid with cash, not with profit, and not with EBITDA.**

PART 1

First principles — what credit risk actually is

1.1 The credit question is not the equity question

An equity investor asks: *"how good can this get?"* Their payoff is unlimited on the upside, so they are paid to imagine success. A lender's payoff is brutally asymmetric: if everything goes perfectly, you receive your coupon and your principal back — nothing more. If things go badly, you can lose most or all of the principal. You participate in none of the upside and all of the downside below your attachment point.

This asymmetry dictates the entire discipline. The credit question is: **"how bad can this get, and do I still get repaid in that world?"** Every ratio, covenant and red flag in this handbook exists to answer that one question. When you review a deal, you are not judging whether the company is good; you are judging whether the company can fail to hit its plan by a wide margin and still return your money on the contractual dates.

1.2 Expected loss: the three quantities that define every credit

$$\text{Expected Loss (EL)} = \text{PD} \times \text{LGD} \times \text{EAD}$$

- **PD — Probability of Default.** How likely is the borrower to miss a contractual payment or breach its way into default over the life of the loan? Financial analysis — the ratios in Part 3, the cash-flow work in Part 4 — is mostly PD analysis. PD is driven by cash-flow adequacy, leverage, liquidity, industry volatility and management quality.
- **LGD — Loss Given Default.** If they do default, what fraction of your exposure do you lose after recoveries? LGD is driven by structure: seniority, security, collateral quality and value, guarantees, jurisdiction, and how much debt ranks ahead of or alongside you. Balance-sheet analysis (asset quality, tangible vs intangible assets) is largely LGD analysis.
- **EAD — Exposure at Default.** How much will actually be outstanding when default happens? For a term loan this is roughly the balance; for revolvers, delayed-draw facilities and guarantees it is larger than today's drawn amount — distressed borrowers draw everything they can before defaulting.

Keep the three separate in your head. A common junior error is to blur them: "the loan is secured, so the credit is fine" confuses low LGD with low PD. A fully secured loan to a business that cannot pay its interest is still a bad loan — you have signed up for a workout, with its costs, delays and uncertainty. Conversely, an unsecured loan to an overwhelmingly cash-generative borrower can be excellent credit. Price and structure must compensate for the *combination*.

1.3 Debt is repaid with cash — the single most important sentence in this handbook

Profit is an accounting opinion; cash is a fact. A company can report growing revenue and positive net income for years while its cash drains away — because revenue is booked when earned (not collected), because costs can be capitalised, and because working capital silently absorbs money. It is entirely possible — and it is precisely the failure your CRO described — for a borrower to show adequate EBITDA while being arithmetically incapable of meeting a principal repayment two years out.

THE LENDER'S ARITHMETIC

For every deal, before anything else, do this sum for each year of the facility's life:

Cash available for debt service = Cash from operations (after working capital and tax) – maintenance capex

Debt service due = cash interest + scheduled amortisation + bullets/maturities falling due that year

If cumulative cash available falls materially short of cumulative debt service — including the bullet — the deal is a **refinancing bet**, and you must underwrite the refinancing (Part 4), not the ratios. Most credit disasters that "surprise" lenders were visible in this two-line calculation at origination.

1.4 The ways out — rank the sources of repayment on every deal

Old-fashioned bank credit training taught the "ways out" hierarchy, and it remains the best single organising frame:

Way out	Source	What you must verify
First	Operating cash flow of the borrower	The FCF walk in Part 4. This should repay (or substantially delever ahead of) the debt in the base <i>and</i> in a plausible downside case. A deal that fails here is not a first-way-out deal, whatever the collateral.
Second	Refinancing or capital-markets take-out	Will leverage, trends and market conditions at maturity support a new lender coming in? This is a bet on a future market. Quantify it (Part 4.4); never assume it.
Third	Support: sponsor/parent equity, guarantees	Is support <i>committed</i> (documented, enforceable) or merely <i>expected</i> (reputational)? Uncommitted sponsor support is worth close to zero in your base case and exactly zero in a fund's year-9 wind-down.
Fourth	Asset sales / collateral enforcement	Realistic distressed value net of prior claims, enforcement cost and time. Enforcement is the admission that everything above failed; it is a recovery estimate, not a repayment plan.

A deal should stand on its first way out, or — where the product is by design refinance- or asset-dependent (real estate, NAV lending, venture debt) — the second/fourth way out must be underwritten with explicit, conservative, evidenced assumptions. Writing "exit via refinancing" without analysis is how the deal in your CRO's example got approved.

1.5 The Five Cs — the qualitative frame around the numbers

C	The question	Where you see it in the financials
Character	Do management and owners behave with integrity and competence? Do they do what they said they would?	Forecast accuracy vs prior years' actuals; aggressive vs conservative accounting choices; related-party transactions; timeliness and quality of information; auditor opinions and changes.
Capacity	Does the business generate enough cash to service the debt? The decisive C.	The cash flow statement; coverage and DSCR ratios; FCF vs debt service schedule.
Capital	How much of someone else's money stands between you and the loss?	Equity cushion, tangible net worth, LTV, sponsor's cash equity (not rolled-over paper gains) at risk beneath you.
Collateral	If it all fails, what do you recover, how fast, and at what cost?	Asset quality and mix on the balance sheet; tangible vs intangible; how assets value in distress; prior-ranking claims.
Conditions	Industry, cyclical position, competitive dynamics, rates, and the terms of the loan itself.	Margin trends vs peers; customer concentration in the notes; sensitivity of demand to the cycle; covenant package and documentation quality (Part 7).

1.6 Think through the cycle, not at the point

Every ratio you compute is a photograph of one moment, usually taken at the best moment (companies raise debt when results look good). Your job is the film, not the photograph. Three disciplines force this:

- **Trend beats level.** A 2.5x interest cover that was 4.0x two years ago is a worse credit than a stable 2.2x. Always compute at least three years of history and ask *what is the direction and what is the mechanism behind it*.
- **Normalise the cycle.** For cyclical borrowers (chemicals, shipping, construction, commodities), leverage on peak earnings is fiction. Ask what mid-cycle and trough EBITDA look like and compute debt capacity on those.
- **The downside case is the underwriting case.** Management's base case is advocacy. Build a downside that is plausible, not apocalyptic — revenue −5–15%, margins −100–300bps, working capital +5–10 days, refinancing +150–250bps — and require the deal to survive it: covenants unbreached or curable, liquidity positive, and a credible path to repayment intact.

FIRST-PRINCIPLES SUMMARY

1. You are paid a few hundred basis points to take a risk of losing 100%. Your analysis exists to make the downside survivable, not to admire the upside.
2. $EL = PD \times LGD \times EAD$ — keep the three separate; price and structure for the combination.
3. Debt is repaid with cash. EBITDA is not cash. Profit is not cash. Only cash is cash.
4. Rank the ways out; a deal should stand on its first way out or the later ways out must be explicitly underwritten.
5. Trend beats level; downside beats base; through-the-cycle beats point-in-time.

PART 2

The three statements through a credit lens

Equity analysts read the statements to estimate value; you read them to find the cash, the claims on the cash, and the lies. Each statement answers a different credit question, and each has characteristic places where problems hide. Read them in this order: **cash flow first, balance sheet second, P&L last** — the reverse of how management presents them, because management presents them in order of how much control they have over the story.

2.1 The cash flow statement — "can they pay me?"

The cash flow statement reconciles accounting profit to actual cash movement, and it is the hardest of the three to manipulate over multi-year periods. Its three sections map directly onto credit questions:

- **Operating (CFO):** does the core business generate cash? Watch the working-capital lines inside CFO — they are where profit and cash diverge. A company can hit its EBITDA covenant while CFO collapses because receivables and inventory are ballooning.
- **Investing (CFI):** what must be reinvested to stay competitive (maintenance capex), and what is discretionary (growth capex, acquisitions)? Get the maintenance/growth split from management and sanity-check it against depreciation — claimed "maintenance capex" persistently below 60–70% of depreciation usually means underinvestment or optimism.
- **Financing (CFF):** how is the gap being plugged? Rising revolver draws, new debt, shareholder loans in — or cash leaking out through dividends, share buybacks and fees to the sponsor.

Free cash flow (lender's definition) = CFO (after tax and working capital) – maintenance capex – cash interest.

This is the cash available to repay *principal*. Compute it yourself; never accept a company's or sponsor's "FCF" line, whose definition will flatter (it commonly excludes working capital, exceptionals, or interest itself).

Cash conversion (CFO ÷ EBITDA) is the fastest single test of earnings quality. A business whose EBITDA is real and whose working capital is controlled converts 85–100%. Persistent conversion below ~70% means EBITDA — the number your covenants, your leverage ratio and your pricing are all based on — systematically overstates cash generation. Find the reason in the working-capital notes before you lend.

2.2 The balance sheet — "what protects me, and what is already owed?"

The balance sheet is a photograph of claims: everything on the right-hand side ranks somewhere relative to you. Read it with four questions:

- **Where does my claim sit?** Map the full capital structure: super-senior RCF, senior secured, second lien, mezzanine, shareholder loans, pensions (often effectively senior), leases, factoring lines (which remove the best receivables from your collateral pool). Structural subordination matters: if you lend to a holding company, operating-company creditors get paid first from operating assets.
- **What are the assets actually worth to a lender?** Haircut mentally: cash (100%, if not trapped abroad or in regulated subsidiaries), receivables (70–85%, less concentration and disputes), inventory (30–60% depending on commodity vs custom), PP&E (site-specific — a bottling line is worth little outside its building), goodwill and capitalised intangibles (≈0 in distress). A balance sheet that is half goodwill — typical after an LBO — means your recovery is the going-concern enterprise value, not assets.
- **What is the working capital telling me?** Compute DSO, DIO, DPO (Part 3.5) for at least three years. Receivables growing faster than revenue, inventory building into slowing sales, payables stretching — each

is a specific, mechanical warning explained in Part 5.

- **What is off the balance sheet or in the footnotes?** Operating commitments, guarantees given, contingent liabilities, litigation, factoring/supply-chain finance (which dresses debt up as payables — ask explicitly), pension deficits, and above all the **debt maturity footnote**: the maturity wall lives there, not on the face of the balance sheet.

2.3 The P&L — "how big is the engine, and is the reported number real?"

The P&L tells you about earnings capacity and trend — necessary, but the most manipulable statement, because it is built on accruals and estimates. Credit reading of a P&L concentrates on four things:

- **Revenue quality.** Growth split into volume vs price; organic vs acquired; recurring/contracted vs transactional; customer concentration (in the notes — one customer above ~15–20% of revenue is a structural risk to be underwritten, not footnoted). Aggressive revenue recognition (bill-and-hold, long-contract percentage-of-completion, big year-end skews) shows up as receivables outrunning revenue.
- **Margin structure and direction.** Gross margin tells you about pricing power vs input costs; EBITDA margin adds overhead discipline. Direction and explanation matter more than the level. A margin that falls while management narrates success is the P&L contradicting the story.
- **The three EBITDAs.** On any leveraged deal there are at least three: **statutory** (from audited accounts), **management-adjusted** (addbacks for exceptionals, synergies, run-rate adjustments), and **covenant EBITDA** (whatever the loan documentation defines — often the most generous of all). Always compute leverage and coverage on all three. The gap between statutory and adjusted is itself a risk metric: above ~20%, you are lending against a spreadsheet, not a company (Part 3.4).
- **Below the line.** Interest expense vs your own calculation of rate × debt (a mismatch reveals capitalised/PIK interest or hidden debt); the tax charge vs cash tax; "exceptional" items that recur (Part 5); capitalised costs (development, software) that convert today's expenses into tomorrow's assets.

2.4 The linkages — where the lies show up

The three statements are one machine, and manipulation in one place must leave a residue somewhere else. The classic checks:

Check	What a mismatch means
Net income + D&A vs CFO, over 3 years ("accruals gap")	Profits persistently not arriving as cash: aggressive revenue recognition, under-provisioning, or structural working-capital absorption. The single best long-range fraud/quality screen.
Revenue growth vs receivables growth	Receivables growing materially faster = channel stuffing, straining terms, or collectability problems.
EBITDA growth vs cash conversion trend	EBITDA up, conversion down = the "growth" is being manufactured in accruals.
Capex vs depreciation, multi-year	Capex ≪ D&A for years = the company is consuming its asset base to show cash flow ("harvesting"); expect a capex catch-up exactly when you least want it.
Interest paid (CF) vs interest expense (P&L)	Expense ≫ paid = PIK or capitalised interest: leverage is compounding out of sight of the cash flow statement.
Tax charge (P&L) vs tax paid (CF), persistent gap	Deferred tax build-up — often a byproduct of aggressive accounting profits that the tax authorities' rules do not recognise.

2.5 Which numbers to trust

- **Audited statutory accounts** are the anchor. Read the audit opinion (qualifications, emphasis of matter, going-concern language), the accounting-policy choices (revenue recognition, capitalisation, provisioning), and the notes — the notes are where the maturity schedule, contingencies, related parties, and customer

concentration live. In private credit you will often receive audited accounts only annually and with a lag: that makes the monthly/quarterly **management accounts** your working material — reconcile them to the last audit and watch for persistent "management vs audited" gaps, which are themselves a character signal.

- **Due-diligence reports** (financial DD by an accounting firm) do the QoE (quality of earnings) work for you: read the EBITDA bridge and the working-capital normalisation with care, and note what the scope *excluded*.
- **Sponsor models** are sales documents. Take the structure, rebuild the assumptions.

PRACTICAL HABIT

Build (or demand) a simple three-statement model for every material exposure — exactly like the companion Excel. If the balance sheet doesn't balance or cash doesn't tie to the cash flow statement, you have found either your own error or theirs; both are worth finding. Every ratio in Part 3 should be a formula tracing to a statement line, never a typed number.

PART 3

The ratio toolkit — every ratio, why it exists, and where it lies

Ratios standardise the statements so that companies of different sizes become comparable and so that trends become visible. Three rules before the toolkit itself. **First: no ratio means anything alone.** Each answers one narrow question; a credit view is the triangulation of all of them plus the cash-flow work in Part 4. **Second: trend beats level** — compute three to five years and explain every inflection. **Third: know each ratio's blind spot.** Every ratio can be gamed; the "traps" column tells you how, and Part 5 lists the resulting red flags. Thresholds given are guides for senior lending to a stable mid-market corporate — calibrate by sector, cyclicity and strategy (Part 6 gives strategy-specific metrics).

3.1 Leverage — how much debt relative to earnings capacity

Ratio	Formula	Guide (senior)	Why it matters / traps
Gross debt / EBITDA	All debt ÷ EBITDA	<4.0x comfortable; 4–5.5x watch; >5.5x aggressive	The market's headline metric: years of earnings to repay debt if every pound of EBITDA went to principal (it never does). <i>Traps:</i> which EBITDA (statutory vs adjusted — always compute both); which debt (include shareholder loans if cash-pay, leases, factoring, pension deficits, earn-outs, PIK accruing).
Net debt / EBITDA	(Debt – cash) ÷ EBITDA	<3.5x / 3.5–5x / >5x	Credits the borrower for cash held. <i>Traps:</i> is the cash real and free — or trapped in foreign subsidiaries, ring-fenced entities, or simply the working-capital float that will be consumed next quarter? Net at year-end can be window-dressed by delaying payables.
Debt / (EBITDA – capex)	Debt ÷ (EBITDA – maintenance capex)	<5.5x / 5.5–7.5x / >7.5x	The honest leverage for capital-intensive businesses: EBITDA before the reinvestment the business <i>must</i> make is not available to creditors. Two companies at 4x EBITDA are entirely different credits if one needs 2% of revenue in capex and the other 8%.
Loan-to-value (LTV)	Net debt ÷ enterprise or asset value	<50% / 50–70% / >70%	Your position in the value stack. <i>Traps:</i> the V. EV = multiple × EBITDA imports every EBITDA problem and adds multiple risk — always sensitise the multiple down 1–2 turns. For asset-based value, use conservative third-party valuations, in distress conditions, net of prior claims.
Debt / total capital (gearing)	Debt ÷ (debt + equity)	<60% / 60–80% / >80%	Book measure of the equity cushion. <i>Traps:</i> book equity is residual accounting — an LBO balance sheet shows "equity" that is mostly capitalised goodwill. Use with tangible net worth (3.7).

3.2 Coverage — can current earnings service the debt

Ratio	Formula	Guide	Why it matters / traps
Interest cover	EBITDA ÷ cash interest	>3.0x; 2–3x watch; <2.0x thin	First-order affordability. Below 2x, more than half of pre-capex earnings goes to coupon — almost no shock absorption. <i>Traps:</i> use <i>cash</i> interest (PIK flatters); use forward rates on floating debt, not last year's average.

(EBITDA – capex) / interest	As stated	>2.0x; 1.3–2x; <1.3x	Cover after mandatory reinvestment — the better test for anything capital-intensive.
DSCR (debt service cover)	$(\text{EBITDA} - \text{capex} - \text{cash tax}) \div (\text{cash interest} + \text{scheduled principal})$	>1.2x; 1.0–1.2x; <1.0x fails	The ratio for "can they pay us back on schedule." Below 1.0x the borrower must borrow, sell assets or burn cash to meet the schedule. <i>Traps</i> : a bullet structure shows a flattering DSCR because scheduled principal is zero — which is exactly why you must also compute the forward DSCR (Part 4).
Fixed-charge cover (FCCR)	$(\text{EBITDA} - \text{capex}) \div (\text{interest} + \text{rents/leases} + \text{scheduled amort.})$	>1.5x; 1.1–1.5x; <1.1x	Adds unavoidable fixed charges — the broadest solvency-in-motion test, standard in ABL documentation. Include lease payments even under IFRS 16 presentation.
Forward / stressed cover	Same ratios at maturity-year forecast, at stressed rates (+150–250bps)	must remain >1.0x DSCR	Cover today is trivia if the debt reprices or matures tomorrow. Part 4 makes this the centre of repayment analysis.

3.3 Liquidity — can they pay what falls due in the next twelve months

Companies die of illiquidity, not of leverage. Leverage kills slowly; a missed payroll kills on Friday. Liquidity analysis is short-horizon and unforgiving:

Ratio	Formula	Guide	Why it matters / traps
Current ratio	Current assets ÷ current liabilities	>1.5; 1.0–1.5; <1.0	Crude solvency-within-a-year screen. <i>Traps</i> : counts inventory at book (often unsaleable in distress) and misses committed undrawn lines. Reclassify structurally-drawn revolvers and any debt inside 12 months as current.
Quick ratio	$(\text{Cash} + \text{receivables}) \div \text{current liabilities}$	>1.0; 0.7–1.0; <0.7	Strips inventory. Better for anything with slow or fashion/obsolescence-prone stock.
Total liquidity	Unrestricted cash + undrawn committed facilities	>10% of revenue, or >6 months of fixed charges	The number that determines survival in a bad quarter. <i>Traps</i> : undrawn lines subject to a borrowing base or a springing covenant may not be drawable exactly when needed; trapped/restricted cash is not liquidity.
Liquidity runway	Total liquidity ÷ monthly cash burn (if FCF-negative)	>12–18 months	The metric for any cash-burning borrower — the core metric in venture debt (Part 6.5).

3.4 Profitability and earnings quality — is the earnings number real

Ratio	Formula	Guide	Why it matters / traps
Gross margin	Gross profit ÷ revenue	trend, vs peers	Pricing power vs input costs — the earliest P&L warning: gross margin erodes quarters before EBITDA does, because overhead cuts can mask the middle of the P&L for a while.
EBITDA margin	EBITDA ÷ revenue	sector-relative	Efficiency of the whole operating model. A borrower with half the peer-median margin has no room for error; a margin far <i>above</i> peers deserves suspicion, not applause — ask what is being capitalised or under-provisioned.
Adjusted-to-statutory gap	$\text{Adjusted EBITDA} \div \text{statutory EBITDA} - 1$	<10%; 10–20%; >20% red	The best single earnings-quality screen in leveraged credit. Every point of gap is leverage hidden from the headline ratio. Studies of LBO vintages repeatedly show projected addbacks/synergies are realised at well under half — underwrite accordingly.

Cash conversion	$\text{CFO} \div \text{EBITDA}$	>85%; 60–85%; <60%	If EBITDA is real it becomes cash. Persistent low conversion = the covenant metric flatters true capacity. Investigate via the working-capital ratios (3.5).
Accruals ratio	$(\text{Net income} - \text{CFO}) \div \text{total assets, 3-yr}$	near 0; persistently >5% red	The academic fraud-screen version of cash conversion. Persistent positive accruals = paper profits.
Return on capital employed	$\text{EBIT} \div (\text{debt} + \text{equity})$	> cost of debt	A business earning less on its capital than it pays for debt is liquidating itself in slow motion. Useful long-lens sanity check on "growth" stories.

3.5 Working capital — where cash hides and earnings are manufactured

DSO = receivables $\div$ revenue $\times$ 365 · **DIO** = inventory $\div$ COGS $\times$ 365 · **DPO** = payables $\div$ COGS $\times$ 365 · **Cash conversion cycle** = DSO + DIO – DPO

- **DSO rising** (say +10 days over two years): customers are paying slower, terms are being stretched to book sales, or doubtful balances are accumulating unprovided. Each day of DSO = revenue/365 of cash absorbed. Cross-check against the bad-debt provision — rising DSO with a *flat* provision means the P&L is not yet admitting what the balance sheet already shows.
- **DIO rising into slowing revenue growth**: demand is fading before the P&L admits it, or stock is obsolete and awaiting a write-down that will hit both EBITDA and your collateral value simultaneously.
- **DPO rising sharply**: flatters cash flow — every extra day is free funding from suppliers — but it is confidence-sensitive credit that snaps back precisely in distress, when suppliers move to cash-on-delivery. A company "generating cash" by stretching payables is borrowing without documentation. Ask about supply-chain finance: it converts debt into innocent-looking payables.
- **The cash conversion cycle** tells you how many days of the trading cycle you are financing. Model working capital as a *consumer* of cash in growth (each £ of revenue growth absorbs CCC/365 pounds of cash) and — counterintuitively — a temporary *source* of cash in decline, which can make a dying company look cash-generative for exactly one year (a notorious trap).
- **Seasonality**: year-end ratios can be window-dressed; ask for monthly working-capital profiles and peak-to-trough facility utilisation.

3.6 Cash-flow ratios — repayment capacity in one line

Ratio	Formula	Guide	Why it matters
FCF after interest	$\text{CFO} - \text{maintenance capex} - \text{cash interest}$	>0 and $\geq$ scheduled amortisation	Cash actually available for principal. Negative with a maturity approaching is the anatomy of your CRO's example deal.
CFO / total debt	As stated	>20%; 10–20%; <10%	Organic payback speed. At 10%, ten years of every operating pound (before capex and interest!) to repay — compare against the actual tenor.
Net debt / FCF ("years to repay")	$\text{Net debt} \div \text{FCF after interest}$	< remaining tenor	If this exceeds tenor, repayment = refinancing or sale by construction. Underwrite that event explicitly (Part 4.4).
Capex / depreciation	As stated	~0.8–1.2 sustainable	<0.7 for years = under-investment (future EBITDA and collateral erosion); persistently >1.3 without growth to show for it = capital indiscipline or capitalised costs.

Shareholder leakage / FCF	$(\text{Dividends} + \text{fees} + \text{buybacks}) \div \text{FCF}$	~0 in stressed credits	Cash leaving to owners while leverage is high is a direct transfer of risk to lenders. Watch sponsor "monitoring fees" and dividend recaps (Part 5).
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3.7 Solvency and structure

Ratio	Formula	Why it matters
Tangible net worth	Equity – goodwill – intangibles	What book cushion exists in assets that survive distress. Negative TNW (standard post-LBO) means recovery = going-concern EV only; your "security" is the assumption that someone will buy the business.
Goodwill / total assets	As stated	>40% flags an acquisition-built balance sheet: impairment risk to covenants that reference net worth, and thin hard-asset recovery.
Secured debt / total assets	As stated, at your ranking	How much of the asset base is already pledged ahead of or alongside you; drives LGD directly.
Contingents & off-BS	Guarantees, litigation, pension deficit, earn-outs, factoring utilisation	Claims that appear only in the notes but rank ahead of or equal to you in a workout. Add them to debt in your own leverage calculation.

HOW TO DEPLOY THE TOOLKIT — THE 30-MINUTE DISCIPLINE

For any new set of financials: (1) compute leverage on statutory *and* adjusted EBITDA, note the gap; (2) interest cover and DSCR, at current *and* stressed rates; (3) cash conversion and the three working-capital day-counts, three-year trend; (4) FCF after interest vs the debt maturity schedule — the Part 4 test; (5) total liquidity vs the next 12 months of obligations. Five clusters, ~20 numbers, and you will have found 90% of what a full committee paper finds. The remaining 10% — documentation, structure, sector — is Parts 6 and 7.

PART 4

Repayment analysis, debt capacity, and the maturity wall

This part is the direct answer to the deal your CRO described: a company approved for credit that was visibly not going to earn enough to meet a repayment falling due within a couple of years. That deal is not caught by any single point-in-time ratio — leverage may pass, interest cover may pass, and a bullet structure makes the standard DSCR look fine because scheduled amortisation is zero. It is caught, every time, by laying **projected cash generation** against the **contractual debt schedule**. This is the core craft of credit analysis; everything in Parts 2 and 3 exists to feed it.

4.1 Build the FCF walk

For each year of the facility life (minimum: to one year past final maturity), from your own conservative rebuild of management's model:

Line	Discipline
EBITDA (lender-adjusted)	Statutory basis; allow only addbacks that are evidenced, non-recurring and cash-verified. No unbanked synergies.
– Working capital movement	From revenue growth × CCC days — growth <i>consumes</i> cash. Add 3–5 days of deterioration in the downside.
– Cash tax	At the real cash rate, not the P&L rate.
– Maintenance capex	Management's split, sanity-checked vs depreciation. Never allow "capex flexibility" to fund debt service in the base case — cutting maintenance capex consumes the collateral.
– Cash interest	At forward curve + margin; PIK accrues to the debt balance instead.
= FCF available for principal	Then: – scheduled amortisation – maturities falling due = surplus / (shortfall) per year , and cumulatively.

Two summary outputs decide the credit:

Organic repayment share = cumulative FCF available over the tenor ÷ total principal due over the tenor.
Forward DSCR at the wall = (EBITDA – capex – tax) in the maturity year ÷ (interest + *all principal maturing that year*).

In the Meridian case study the organic repayment share is ~2% and the forward DSCR at the 2027 wall is ~0.1x. Written that way, no committee approves the deal unless the refinancing itself is underwritten — which is the point. Ratios describe; this calculation decides.

4.2 Debt capacity — how much debt can this business carry?

Reverse the walk. Instead of asking "can they carry the proposed debt," ask "what debt could they carry?" — it anchors negotiation and reveals how much of the ask is hope:

- **Coverage-based capacity:** the debt level at which stressed-case EBITDA still covers interest $\geq 2.0\times$ at stressed rates. (Meridian downside EBITDA ~£26m, stressed all-in rate ~11% → supportable debt $\approx 26/(2.0 \times 0.11) \approx \text{£118m}$ — versus £190m proposed.)
- **Amortisation-based capacity:** the debt fully repayable from downside FCF over a realistic tenor (5–7 years). This is the true "first-way-out" capacity and is usually the most conservative number.

- **Value-based capacity:** stressed enterprise/collateral value $\times$ maximum acceptable LTV. This caps LGD rather than PD.

Debt capacity is the *lowest* of the three unless the strategy explicitly relies on a different way out (e.g. NAV lending is value-based by design — Part 6.3). The gap between capacity and the proposed quantum must be filled by structure: equity, amortisation, sweeps, or a smaller hold.

4.3 The maturity wall

Plot every liability by contractual maturity — term debt, RCF, leases, factoring lines, shareholder instruments with maturity dates, large earn-outs. You are looking for:

- **Concentration:** a single year (or, as with Meridian, a six-month window) holding most of the stack. No sequencing means no early warning and no second chance.
- **Ordering:** does the RCF — the liquidity lifeline — mature *before* the term debt? A revolver that lapses six months before a bullet removes the liquidity buffer exactly when the refinancing drama begins. (Meridian: RCF Mar-2027, bullet Sep-2027 — textbook.)
- **Springing maturities:** documentation that accelerates one instrument if another isn't refinanced by a set date — walls hiding inside walls, found only by reading the docs.
- **Refinancing runway:** a refinancing takes 6–12 months to execute in good conditions. A borrower entering the final 18 months before a wall with no process running is already late; that is a monitoring trigger (Part 9), not merely an observation.

4.4 Underwriting refinancing risk — because sometimes the second way out is the deal

Bullet structures are legitimate; pretending they are not refinancing bets is what is illegitimate. If repayment depends on a future refinancing, underwrite that future transaction as if you were the future lender:

1. **Leverage at the wall, on statutory EBITDA, in your downside case.** Will a rational new lender advance that multiple, in that sector, on those trends? Remember the future lender applies *their* haircuts to addbacks, not the sponsor's.
2. **Coverage at the wall, at stressed future rates** (forward curve +150–250bps). If cover at refinancing pricing is below $\sim 2\times$, the refinancing produces an immediately stressed credit even if it closes.
3. **Trend into the wall.** Lenders refinance improving stories and flee deteriorating ones. Meridian arrives at its wall with falling margins, rising DSO and a missed synergy record: the trend itself is the refi risk.
4. **Market dependency.** Private credit and syndicated markets close, hard, roughly once a decade and briefly more often (2008–09, 2020, 2022–23). A wall inside 2–3 years is a bet that the window is open in a specific half-year. You cannot predict it; you can refuse structures that require it, or price and size for it.
5. **The fallback.** If the refinancing fails: what does the amend-and-extend look like, what would you demand, what does the sponsor contribute, and what is recovery if the answer is nothing? Write it down at origination — it is the pre-mortem that becomes the workout plan.

4.5 Stress testing — the downside case is the underwriting case

- **Standard downside (apply to every deal):** revenue -5% to -15% (calibrated to sector beta), gross margin -100 to -300 bps, working capital $+5$ to $+10$ days, capex at maintenance floor, rates at forward $+200$ bps, refinancing margin $+200$ bps. The deal must show: covenants unbreached or curable, liquidity never below zero (with a buffer), and a path to repayment that survives.

- **Reverse stress test:** find the EBITDA level at which (a) liquidity runs out and (b) the refinancing becomes impossible (leverage > market maximum). Express as "% decline from today." If a routine recession-sized decline (15–25% EBITDA for a cyclical) breaks the deal, the structure is wrong regardless of the base case.
- **Single-point sensitivities:** the biggest customer leaves; the key input cost +20%; FX moves 10%; one covenant of headroom. Sensitivities locate *which* assumption carries the deal — that assumption is where diligence and covenants should concentrate.

HOW THE CRO'S EXAMPLE DEAL GETS CAUGHT — THE CHECKLIST

1. FCF walk to one year past maturity, on lender-adjusted numbers. → cumulative FCF vs total principal due: **organic repayment share**.
 2. **Forward DSCR including maturing principal** in the maturity year — the bullet-blind standard DSCR is not enough.
 3. Maturity wall chart including RCF ordering and springing maturities.
 4. If organic share is low: full refinancing underwrite (leverage/coverage/trend/market at the wall) in the downside case.
 5. Structure the gap: amortisation + cash sweep to force deleveraging before the wall; maturity long enough to give two refinancing windows, not one; covenants that hand you the keys while there is still value.
- Any deal where this page cannot be completed convincingly is a decline, whatever the leverage ratio says.

PART 5

The red-flag catalogue — forty-one warning signs in the financial data

A red flag is rarely proof of a problem; it is a **question the numbers are asking that management has not answered**. The discipline is threefold: (1) find flags mechanically — run these checks on every set of financials, so that finding them does not depend on mood or time pressure; (2) never accept the first explanation — verify against the data ("DSO rose because of one large customer" → show me the ageing report); (3) weigh flags in combination — one flag is a question, three related flags are a pattern, five are a decision. Flags marked ♦ are the ones that most directly produce the "cannot repay the upcoming maturity" failure.

5.1 P&L red flags

Flag	Why it predicts credit loss
Recurring "exceptional" or "one-off" costs (2+ consecutive years)	They are operating costs being laundered out of covenant EBITDA. Leverage and coverage are overstated by the full amount, every year.
Large addbacks: unrealised synergies, "run-rate" adjustments, pro-forma acquisitions	Lending against projected EBITDA that has never existed. Realisation rates on projected synergies are chronically poor; every point of addback is hidden leverage discovered at refinancing.
Adjusted-to-statutory EBITDA gap > 20%	The composite of the two above — the best single earnings-quality screen.
Gross margin eroding while management narrates strength	No pricing power against input costs; the earliest P&L warning, visible quarters before EBITDA cracks.
Revenue growth from price only, volumes flat/falling (or unexplained mix)	Inflation-flattered growth that reverses when input costs turn; volume decline is the real demand signal.
Revenue growth decelerating sharply vs plan, year after year	The market is voting against the story; forecast credibility falls with each miss.
Interest expense ≠ your calc of rate × debt	PIK, capitalised interest or undisclosed debt. Compounding leverage out of sight.
Capitalising costs peers expense (development, software, "customer acquisition")	Today's costs become tomorrow's assets: EBITDA and equity both inflated; the write-down arrives with the downturn.
Hockey-stick forecasts ahead of a financing event	Advocacy, not analysis. Score management on last year's forecast vs actuals instead.
P&L tax charge persistently » cash tax paid	Aggressive accounting profits the tax rules don't recognise; deferred tax building.

5.2 Balance-sheet red flags

Flag	Why it predicts credit loss
♦ Receivables growing materially faster than revenue (DSO +10d)	Channel stuffing, stretched terms to book sales, or collectability rot. Revenue quality degrading; a provision hit is queued.
♦ Inventory building into slowing sales (DIO rising)	Demand fading before the P&L admits it; obsolete stock awaiting write-down — hits EBITDA and collateral simultaneously.
♦ Payables stretched sharply (DPO +10d)	Borrowing from suppliers without documentation; snaps back to cash-on-delivery at the first wobble — a liquidity cliff.
Supply-chain finance / factoring appearing or growing	Debt disguised as payables; your best collateral (receivables) sold out of the pool. Ask explicitly — disclosure is often minimal.

◆ Cash falling while revenue grows	Growth consuming rather than generating cash; the balance sheet is being liquidated into the P&L.
◆ Revolver structurally drawn, utilisation creeping up	The emergency buffer is spent on business-as-usual; no shock absorber remains.
Goodwill > 40% of assets / negative tangible net worth	Recovery depends entirely on going-concern EV; impairment risk to any net-worth covenant.
Debt creeping up in "small" increments (leases, local lines, earn-outs)	Leverage rising between covenant tests; each item individually immaterial, collectively a turn of EBITDA.
Pension deficit, litigation reserves, guarantees in the notes	Senior-ranking or pari claims not in the headline leverage number. Add them yourself.
Trapped cash (minority JVs, foreign subs, regulated entities)	"Net debt" flatters; the cash cannot reach the debt it nominally nets against.
Asset revaluations / fair-value gains boosting equity	Paper cushion. In distress the revaluation reverses exactly when the covenant needs it.

5.3 Cash-flow red flags

Flag	Why it predicts credit loss
◆ FCF after interest ≤ 0 with a maturity inside 3 years	The anatomy of your CRO's example. No organic repayment capacity + a contractual repayment = pure refinancing bet. Part 4 is the response.
◆ Cash conversion (CFO/EBITDA) persistently < 70%	The covenant metric overstates cash earnings; find where the cash is leaking (usually 5.2's flags).
Net income positive, CFO weak, 3+ years (accruals gap)	Long-range earnings-manipulation screen; paper profits.
CFO propped by one-offs: tax refunds, disposals in "operating", payables stretch	Non-repeatable cash dressed as operating strength — normalise it out.
Capex $\ll$ depreciation for 2+ years	Harvesting: consuming the asset base to show FCF. Expect a forced capex catch-up and eroded collateral.
Dividends / sponsor fees / buybacks while leverage is high or rising	Cash exiting to owners = risk transferring to lenders. A dividend recap resets the deal — re-underwrite from scratch.
"Growth capex" that never produces revenue growth	Either it is really maintenance (EBITDA-capex leverage understated) or capital is being destroyed.
Working capital as a cash <i>source</i> in a shrinking business	Mechanical one-year effect of decline that masks operational collapse — the dying company that "generates cash" for exactly one year.

5.4 Ratio-trend and structural red flags

Flag	Why it predicts credit loss
◆ Concentrated maturity wall (most of the stack in one window)	One closed market window = default regardless of operating performance.
◆ RCF maturing before the term debt	Liquidity lifeline lapses exactly when refinancing stress begins.
Leverage flat/rising despite "strong performance" narrative	The narrative and the balance sheet disagree; believe the balance sheet.
Covenant headroom shrinking quarter over quarter	Trajectory toward breach; act on the trend, not the breach — by breach the options are worse.
Repeated covenant resets / amendments / equity cures	Each amendment is a default that was negotiated instead of declared.
Ratios fine on adjusted EBITDA, failing on statutory	The entire credit case rests on the addbacks — see 5.1.

5.5 Qualitative and disclosure red flags (visible from the finance pack)

Flag	Why it predicts credit loss
Auditor resignation/change, especially downward-tier; qualified opinions; emphasis of matter; going-concern language	The people with the most access are distancing themselves. Treat any audit qualification as a credit event until explained.
CFO or FD departure, particularly ahead of results or refinancing	The person who knows the numbers best has left the building.
Late, restated, or repeatedly delayed accounts; deteriorating information quality	Either the systems are failing or someone needs the extra time. Both are credit problems. Information quality is a covenant — enforce it.
Related-party transactions: sales to affiliates, loans to directors, assets moving to sister entities	Value and collateral can leak to entities outside your security net. J.Crew-style asset transfers start here (Part 7).
Change of accounting policies that flatters (revenue recognition, capitalisation, provisioning, depreciation lives)	Same business, better optics — the change itself is the signal; restate the trend on the old policy.
Customer/supplier concentration worsening in the notes	One counterparty decision can remove a quarter of EBITDA; concentration is leverage you cannot see in debt ratios.

USING THE CATALOGUE

Run 5.1–5.5 as a literal checklist at origination and at every annual review. Record each flag found, the explanation received, and the evidence verifying the explanation. Three related flags = pattern: escalate to watchlist analysis (Part 9) even if every covenant is green. The Meridian case in the Excel contains fifteen of these flags — find them yourself before reading the workbook's Red Flags sheet.

PART 6

Strategy by strategy, A to Z

Parts 1–5 apply everywhere. But each private credit strategy has a different **repayment engine**, and therefore a different analytical centre of gravity. The fatal error is analysing every deal like a corporate loan: a subscription line, a NAV facility and a venture loan can all look "unleveraged" through a corporate-ratio lens while carrying serious risk in dimensions corporate ratios never touch. For each strategy below: what it is, what repays you, the analysis step-by-step, the metrics with guide levels, the structure that protects you, and the strategy-specific red flags.

6.1 Direct lending (sponsor-backed and sponsorless corporate)

What it is. Senior secured (unitranche, first-lien, occasionally second-lien/mezz) loans to mid-market companies, typically PE-owned, 3–7 year tenors, floating rate, bullet or minimal amortisation. This is the core of the asset class, and the Meridian case study is a direct-lending deal.

What repays you. First way out: borrower FCF (rarely sufficient for the bullet); in practice: refinancing or exit (sale/IPO) by the sponsor. Direct lending is therefore structurally a refinancing/exit-dependent product — which makes Part 4 the centre of the analysis, not an appendix.

The analysis, step by step.

- Business quality:** market position, customer concentration, contract vs transactional revenue, cyclicality, pricing power (gross-margin history through the last downturn is the evidence).
- Earnings quality:** rebuild EBITDA from statutory; haircut addbacks (a common house rule: cap accepted addbacks at 10–15% of statutory, take 0–50% of "synergies" depending on evidence); read the QoE report's working-capital normalisation.
- Full ratio pass** (Part 3) on statutory, adjusted, and covenant EBITDA.
- FCF walk and refinancing underwrite** (Part 4), base and downside.
- Sponsor assessment:** track record in the sector, behaviour in past workouts (do they support or hand over keys?), fresh cash equity % beneath you (30–50% of the structure is market norm — less means you are the equity), fund vintage (a year-9 fund has no dry powder and no patience).
- Structure and docs** (Part 7): covenants, EBITDA definition, leakage baskets, security perimeter.
- Relative value:** is the spread paying for the risk vs comparable deals? Risk you are not paid for is risk you should not hold.

Key metrics	Guide (senior mid-market)	Notes
Net debt / EBITDA (statutory)	4.0–5.5x typical; >6x aggressive	Always alongside the adjusted-gap metric.
EBITDA / cash interest	≥2.0x at underwriting, at forward rates	Floating-rate books: re-test at +200bps.
Equity cushion (sponsor cash equity / EV)	≥35–40%	At entry multiple; sensitise the multiple down.
LTV through your tranche	≤55–60%	On a defensible EV, not the deal EV.
Organic repayment share (Part 4)	state it on every deal	Low is normal for bullets — but it must be <i>stated</i> , so the refi bet is priced consciously.

Red flags specific to the strategy: purchase multiple far above sector norm (your "equity cushion" is a valuation opinion); dividend recap inside 18 months of close; sponsor rolling over paper gains rather than fresh

cash; covenant-lite with a wide-open EBITDA definition at mid-market size (large-cap terms without large-cap liquidity); portfolio-company fees streaming to the sponsor while leverage rises.

6.2 Fund finance — subscription / capital-call facilities (RCF)

What it is. A revolving facility to a closed-end fund (PE, PC, infra, RE), secured not on portfolio assets but on the **unfunded capital commitments of the fund's investors (LPs)** — the lender takes security over the right to call capital and over the account into which calls are paid. Used to bridge capital calls, smooth drawdowns, and (controversially) enhance reported IRR by delaying calls. Tenor typically 364 days–3 years, utilisation fluctuating.

What repays you. Capital calls on LPs. You are not underwriting the fund's investments at all in the first instance — **you are underwriting the LPs' willingness and ability to fund calls**, and the enforceability of your right to force those calls. This is the strategy where corporate ratio analysis is most useless and where the "financial data" to analyse is the LP register and the fund's limited partnership agreement (LPA).

The analysis, step by step.

1. **Borrowing base:** which LPs count ("included investors"), at what advance rates? Typical: rated institutional LPs 90%; strong unrated institutions 65–80%; HNW/feeder/retail vehicles 0–50%. The facility limit = advance rate × eligible unfunded commitments.
2. **LP quality and concentration:** ratings/financial strength of the top LPs; single-LP concentration caps (e.g. largest ≤15–20% of the base); granularity; history of any LP defaulting on calls (rare — sub-1% historically — but the entire product rests on it staying rare); overlap risk (the same LPs across many funds you lend to = correlated exposure across your book).
3. **Documentation/enforceability:** does the LPA explicitly permit the pledge of uncalled commitments? Can LPs' obligations be excused (sovereign immunity, ERISA, "excuse rights")? Are side letters eroding the call obligation? Can the GP be removed — and does your security survive GP removal?
4. **Fund lifecycle:** uncalled commitments only fall as the fund deploys. A sub-line late in the investment period is secured by a shrinking base — monitor the coverage ratio (unfunded commitments ÷ facility) continuously, require clean-downs, and hard-stop the facility before the investment period ends.
5. **Purpose discipline:** bridging calls (fine) vs funding at-risk investments for extended periods or masking LP liquidity problems (not fine). Long-undrawn-then-permanently-drawn utilisation patterns are the tell.

Key metrics	Guide	Notes
Coverage: eligible unfunded commitments / commitments drawn	≥2.0x weighted	Recompute as the fund deploys; the base only shrinks.
Advance rate achieved	≤65% blended	Against the borrowing-base grid, after concentration caps.
Largest LP / included base	≤15–20%	And know who they are across your whole fund-finance book.
Clean-down frequency	≥1x/year to zero (or per docs)	Proves the line is bridging, not structural leverage.
% of investment period remaining	facility maturity well inside it	Plus mandatory reduction as unfunded burns down.

Red flags: retail/semi-liquid feeder LPs in the base (redemption runs can hollow out commitments); GP requesting extensions while utilisation stays at cap; NAV of the fund falling sharply (LPs facing losses are slower to fund — and "excuse" litigation becomes thinkable); side letters you have not read; a GP under regulatory or key-person stress (call enforceability gets tested exactly when the GP is weakest); facility being used to delay calls for IRR optics into a falling market.

6.3 NAV financing

What it is. A loan to a fund (or holdco) secured on the **net asset value of its portfolio** — the inverse of a sub-line: late-lifecycle, when commitments are drawn and the fund holds assets. Used for follow-ons, distributions to LPs (scrutinise this), or bridging exits. LTVs are low (5–25% typical for diversified PE portfolios) precisely because the collateral is illiquid, hard to value, and cross-correlated.

What repays you. Distributions from portfolio realisations (exits) cascading to you ahead of LPs. Your analysis is therefore: **portfolio quality × diversification × valuation credibility × exit timing**, wrapped in a covenant that forces deleveraging if value falls.

The analysis, step by step.

- 1. Look through to the portfolio:** how many assets, what sectors/vintages/geographies, single-asset concentration (largest $\leq 15\text{--}20\%$ of NAV for a diversified facility). A "diversified" portfolio of 8 assets where 3 are 60% of NAV is a concentrated facility priced as a diversified one. For each major asset: leverage *inside* the asset (your NAV collateral is equity value — already levered once), performance vs plan, realistic exit route and timing.
- 2. Interrogate the NAV itself:** who values, how often, audited by whom? GP marks lag public markets by 2–3 quarters in a downturn. Compare recent exits vs their prior carrying marks (exits consistently above marks = conservative GP; the reverse is disqualifying). Haircut NAV yourself: apply a stress (–20–40% depending on portfolio) and check the covenant cascade still protects you.
- 3. Model the distribution waterfall:** when do exits realistically occur (add 12–24 months to GP guidance in your downside); what ranks ahead (asset-level debt, fund expenses, GP clawbacks); does your cash sweep capture distributions before LPs are paid?
- 4. Covenant cascade:** LTV triggers (e.g. cash sweep at 25%, cure/margin call at 30%, default at 35–40%) referencing an *independently verifiable* NAV; valuation dispute mechanics; portfolio concentration covenants.
- 5. Alignment:** why does the GP want the money? Follow-on capital into performing assets is a good reason. Funding LP distributions from a fund that cannot exit assets is leverage replacing liquidity — the distribution the facility funds is value leaving your collateral pool on day one.

Key metrics	Guide	Notes
LTV (loan / eligible NAV)	$\leq 15\text{--}25\%$ diversified PE; lower if concentrated	On your own stressed NAV, not the GP's marks.
Largest asset / NAV; top-3 / NAV	$\leq 20\%$; $\leq 45\%$	Concentration is the dominant risk driver.
Look-through leverage	compute it, always	Facility + weighted asset-level debt / gross asset value: the true attachment point.
Realisations coverage	expected distributions over tenor $\geq 2\times$ debt service + bullet	On sale timings you have stressed.
Marks vs exits track record	exits $\geq$ carrying marks	The single best test of NAV credibility.

Red flags: facility proceeds paying LP distributions late in fund life; NAV flat while comparable public markets fell 25% (stale marks); concentration creeping as assets exit and the loan doesn't amortise (the good assets leave first — require distributions to prepay); valuation policy changed or valuer switched; GP fundraising for the next vintage stalling (removes the incentive to protect track record); asset-level lenders tightening (your collateral's equity being squeezed from below).

6.4 Trade finance

What it is. Short-tenor (30–180 day), self-liquidating financing of specific trade flows: receivables finance/factoring, borrowing-base facilities on inventory and receivables, pre-export and prepayment finance, letter-of-credit confirmation and discounting. The unit of risk is the **transaction**, not the company: the loan is designed to repay from a specific, identifiable cash flow (the buyer paying for the goods).

What repays you. The completion of the underlying trade. Analysis therefore splits into: (a) **performance risk** — will the goods exist, ship, and be accepted; (b) **payment risk** — will the obligor (buyer) pay; (c) **structure integrity** — does the cash actually flow through channels you control; (d) **fraud** — the dominant realised loss driver in this strategy. Corporate ratios matter mostly for the *seller's* ability to keep performing and for dilution risk.

The analysis, step by step.

1. **Obligor credit:** the buyer is your real credit in receivables deals — full Part 3 analysis where financials exist, plus payment-behaviour history (days-beyond-terms trend), credit insurance availability and wording (does the policy actually pay on protracted default, and can you claim directly?).
2. **Dilution analysis (receivables):** historical dilution — returns, disputes, rebates, offsets, credit notes — sets the advance rate. Typical: 80–90% advance on investment-grade obligor books, less concentration > caps, less dilution reserve at 1.5–2× historical dilution.
3. **Transaction integrity:** verify the trade is real: purchase orders, bills of lading, warehouse receipts (from independent, inspected warehouses), inspection certificates. **Duplicate-financing risk is the classic fraud** — the same invoice or warehouse receipt financed with three lenders. Mitigants: direct obligor verification of invoices, assignment notices, registry checks, control of original documents.
4. **Cash-flow control:** collections into a pledged/assigned account you control; true sale vs security assignment (insolvency treatment differs); commingling risk if collections route via the seller.
5. **Concentration and tenor discipline:** obligor, country and commodity caps; tenor matched to the actual trade cycle — a "90-day" facility that keeps rolling the same exposure is a term loan wearing a trade-finance costume (and pricing).
6. **Seller/borrower financial health** (Part 3, light version): a failing seller produces disputes, dilution and fraud incentives even when obligors are sound. Watch seller DSO/DPO and liquidity.

Key metrics	Guide	Notes
Advance rate vs (100% – dilution reserve – concentration excess)	≤85–90% net	Dilution reserve ≥1.5–2× trailing-12m dilution.
Historical dilution	<3–5% stable	Rising dilution = product/relationship problems upstream of credit.
Obligor concentration	largest ≤10–20% of pool	Higher only with credit insurance you have verified.
Days-beyond-terms trend	stable	Earliest payment-behaviour warning, monthly.
Roll-over rate / true turnover of the book	exposures actually self-liquidate	Ever-greening is the structural red flag.

Red flags: invoices in round numbers, or to obligors you cannot independently verify; the same collateral appearing across lenders (registry checks!); warehouse receipts from affiliated or unvisited warehouses; sudden growth in "trade" volume outrunning the seller's operational capacity; obligor confirmations refused or routed through the seller; requests to extend tenor mid-transaction ("the buyer needs more time"); commodity-price moves leaving transactions underwater mid-flight; collections slowing into the seller's own liquidity crunch (temptation to divert your cash).

6.5 Venture debt

What it is. Term loans (typically 3–4 years, amortising after an interest-only period, plus warrants) to VC-backed, usually loss-making growth companies, sized at 20–35% of the last equity round. The borrower has negative EBITDA by design: **every corporate coverage ratio in Part 3 fails at origination.** That is not a red flag here; it is the product. The analysis has a different engine.

What repays you. In order of reality: the **next equity round** (the dominant true source), acquisition of the company, or — for the best cases — the company reaching cash-flow breakeven. You are underwriting *the probability that credible investors write the next cheque before the cash runs out*, with enterprise value (usually via an all-assets lien and sometimes IP security) as the recovery backstop.

The analysis, step by step.

- Runway arithmetic (the centre of the file):** months of runway = unrestricted cash ÷ current net monthly burn. Rules: runway at close ≥ 12–18 months *after* your loan is drawn; your final maturity and amortisation start must sit *inside* the funded runway or behind a credible milestone; never be the money that extends a runway to a milestone the equity investors declined to fund.
- Burn quality:** split burn into gross burn (all cash costs) and net burn (after revenue). Is burn producing growth (efficient) or just existing (terminal)? Metrics: burn multiple = net burn ÷ net new ARR (<1.5 good, >2.5 alarming); months to breakeven at current trajectory vs runway.
- Revenue quality (SaaS/recurring lens):** ARR growth rate; gross/net revenue retention (NRR >100% = the installed base grows by itself; GRR <85% = the bucket leaks); gross margin (real software ~70–85%; "software" at 40% is a services business priced as software); customer concentration; magic number / CAC payback for sales efficiency.
- Investor syndicate strength — your real repayment source:** who is on the cap table, do they have reserves and dry powder, did insiders lead or bridge the last round (insider-led flat/down rounds = external validation failing), what is the explicit reserve policy for this company? Talk to the investors: "will you fund the next round?" is a question you are allowed to ask, and the hesitation is data.
- Enterprise-value cushion:** last-round post-money vs your debt (loan ≤ 20–35% of last round); but mark the round to current market conditions — a 2021-vintage valuation is not collateral in 2026. Acquisition comparables for the technology/team as recovery floor; IP lien where it has standalone value.
- Structure:** interest-only period matched to milestones; MAC/investor-abandonment triggers; minimum-cash/minimum-runway covenants (the venture-debt equivalent of maintenance covenants); warrants for the risk you are actually taking.

Key metrics	Guide	Notes
Runway post-draw	≥12–18 months	Recompute monthly from the MIS; the covenant that matters.
Loan / last equity round	≤20–35%	And ≤ ~1 × ARR for SaaS as a cross-check.
Burn multiple (net burn / net new ARR)	<1.5 good; >2.5 red	Growth efficiency — is the burn buying anything?
Net revenue retention	>100%	<90% means growth is a treadmill against churn.
Gross margin	>70% (software)	Validates the business model the valuation assumes.

Red flags: runway shortening faster than plan (burn discipline failing); insider-led or structured (liquidation-preference-heavy) rounds; flat/down round or round repeatedly delayed ("we'll raise in Q3" for three consecutive quarters); NRR sliding below 100%; founders or key engineers departing; the company drawing your facility in full immediately at close (you are the runway); receivables building in an "ARR" business (annual prepay becoming monthly, or worse, non-payment); lead investor marking the position down or going quiet in board minutes you are entitled to see.

THE ONE-LINE SUMMARY OF PART 6

Direct lending: underwrite the **cash flow and the refinancing**. Sub-lines: underwrite the **LPs and the call mechanics**. NAV: underwrite the **portfolio, the marks and the waterfall**. Trade finance: underwrite the **transaction, the obligor and the fraud surface**. Venture debt: underwrite the **runway and the next round**. Same first principles, five different repayment engines — always name the engine before you open the spreadsheet.

PART 7

Covenants, documentation and structural protection

Analysis tells you whether to lend; structure determines what happens when the analysis turns out to be wrong. A well-structured mediocre credit often outperforms a badly structured good one, because structure controls *when you find out* and *what you can do about it*. Documentation is credit analysis continued by other means — and in current markets it is where most risk quietly enters.

7.1 Covenants — the early-warning and control system

- **Maintenance covenants** are tested every quarter regardless of borrower actions: net leverage $\leq X$ (usually stepping down), interest/fixed-charge cover $\geq Y$, sometimes minimum liquidity or minimum EBITDA (venture debt: minimum cash/runway). They are your smoke alarm: a breach forces the borrower to the table *while the company still has value*. Set at 25–35% headroom to the base case: wider is not "borrower-friendly", it is an alarm wired to go off after the fire.
- **Incurrence covenants** test only when the borrower acts (new debt, dividends, acquisitions). A cov-lite loan has only these: you have no seat at the table until money is leaving or the borrower defaults on payment — by which time the good outcomes have usually left too. Cov-lite is a market reality in larger deals; in the mid-market, insist on at least one maintenance leverage covenant. Price cov-lite as what it is: the same PD, later detection, worse LGD.
- **Information covenants are covenants.** Monthly management accounts (30 days), quarterly covenant certificates with computations shown, annual audited accounts (120–150 days), annual budget, notification of material events. Late or degrading information is itself an early-warning indicator (Part 5.5) — enforce the timeline.
- **Equity cure rights:** limit number (e.g. 2–4 over the life, not consecutive), require cash actually contributed and applied to deleveraging, prevent "over-cures" that reset the covenant with paper.

7.2 The EBITDA definition — the covenant behind the covenant

Every ratio covenant is only as strong as its EBITDA definition, and this is where modern documents are engineered. Read, in every deal, the definition of "Consolidated EBITDA" and hunt for: **uncapped addbacks** for synergies and cost savings "expected to be realised" (demand a cap $\leq 10\text{--}20\%$ of EBITDA and a time limit $\leq 12\text{--}18$ months); **pro-forma adjustments** at the borrower's discretion; **run-rate revenue** from signed-but-unbilled contracts; and **"Grower" baskets** (the greater of a fixed amount and % of EBITDA) that inflate every permission as the inflated EBITDA grows. A borrower with a 5.0x covenant and an open EBITDA definition effectively has no covenant: the leverage test is being marked by the student.

7.3 Leakage — keeping the value inside your security net

- **Restricted payments and permitted investments baskets:** how much cash/assets can leave to shareholders or unrestricted subsidiaries? Sum every basket and assume full use on day one — that sum is your real cushion, not the headline equity.
- **Unrestricted subsidiaries and asset transfers:** the J.Crew manoeuvre — moving crown-jewel IP/assets into an entity outside the security net, then borrowing against it. Blockers: no material IP/asset transfers to unrestricted subsidiaries, "no worse position" tests, designation caps.
- **Priming and uptiering:** the Serta manoeuvre — a majority lender group amends the docs to insert new super-priority debt ahead of the minority. Blockers: sacred-right protections on subordination requiring all-lender consent, pro-rata sharing that cannot be amended by simple majority.

- **Guarantor coverage tests:** $\geq 80\text{--}85\%$ of group EBITDA and assets inside the guarantor/security group, tested annually — otherwise the group grows out of your security perimeter.
- **Change of control, sweeps, and MFN** on incremental debt complete the standard checklist.

7.4 Structure as compensation for analysis risk

Where Part 4 shows weak organic repayment, structure must close the gap: **amortisation + excess-cash-flow sweep** (typically 50%, stepping down with leverage) forces deleveraging before the wall; **shorter tenor or springing maturity** keeps you ahead of other creditors; **super-senior or first-out positioning** buys LGD when PD cannot be improved; **pricing** (margin, OID, call protection) is the last resort — spread never repairs a broken repayment path, it only pays you while you wait to find out.

PART 8

The worked case study — Meridian Foods, end to end

Open **Credit_Analysis_Worked_Example.xlsx** alongside this chapter. Meridian Foods Group Ltd is fictional, but it is engineered from the standard anatomy of real private-credit losses — and specifically the failure your CRO described: a deal approved while the borrower was visibly not going to earn enough to meet a repayment two years out. Work the sheets in the order below; the workbook's README sets the same drill.

8.1 The situation

UK private-label food manufacturer, revenue £248m, PE-owned since 2022. Facilities: **£175m unitranche, bullet Sep-2027, SONIA+600; £15m super-senior RCF, Mar-2027**, £10m drawn. The sponsor quotes leverage of "4.2x adjusted EBITDA" and points to three consecutive years of revenue growth. The ask: hold the position / roll at refinancing.

8.2 What the P&L says (PnL sheet)

- Revenue growth is real but decelerating: +10.5% → +6.9%, forecast ~3–4%. Gross margin has fallen 250bps in two years (COGS 68.0% → 70.5%): input inflation is not being passed through — a pricing-power problem in a private-label business selling to concentrated grocers.
- "Exceptional" restructuring costs appear in all three historical years (£2.0/3.5/4.0m) and are still forecast — they are operating costs. Management Adjusted EBITDA of £42m vs statutory £31m: a **35% adjustment gap**, mostly recurring "one-offs" plus £3m of synergies that two years of actuals have never evidenced.
- Consequence: quoted leverage 4.2x (adjusted) is really **6.0x gross / 5.8x net on statutory EBITDA**. Interest of £18.7m consumes 60% of statutory EBITDA — cover 1.66x. PAT is ~£1.7m: the equity cushion rebuilds at £1–2m a year beneath £189m of debt.

8.3 What the balance sheet says (BS sheet)

- DSO 48 → 63 days (receivables +55% vs revenue +18%); DIO 55 → 60; DPO stretched 52 → 66. The cash conversion cycle looks stable *only because suppliers are funding the deterioration* — confidence-sensitive funding that snaps back in distress.
- Cash has fallen £12.0m → £6.5m while the RCF crept £0 → £10m of £15m: total liquidity is £11.5m (4.6% of revenue) and shrinking. The buffer is already spent.
- Goodwill £145m = 47% of assets; tangible net worth **–£82m**. Recovery = going-concern EV or nothing.

8.4 What the cash flow statement says (CF sheet) — the verdict

- Cash conversion ~72% and falling: working capital has absorbed ~£18m over three years — more than one full year of post-interest earnings.
- FCF after interest: FY25 = –£2.5m**. Forecast (on management's own hockey-stick) ~£1.0m and £2.9m in FY26–27. Cumulative cash available before the bullet: **<£4m against £175m due** — an organic repayment share of ~2%.

8.5 What the debt schedule says (Debt sheet)

- £189m — the entire stack — matures in a six-month window in 2027, with the RCF lapsing *first* (Mar-2027), removing the liquidity line exactly as refinancing pressure peaks.
- Refinancing test at the wall: funding gap $\approx$ £176m; leverage at refinancing $\approx$ 4.7x statutory (before the new lender's own addback haircuts); interest cover if refinanced at +200bps $\approx$ 1.6x. The refinancing case requires a benign market, a lender who accepts deteriorating trends, and synergies finally arriving — simultaneously.

8.6 The downside case (always run one)

Revenue -5%, gross margin -100bps further, DSO +5 days, refinancing at +200bps: EBITDA $\sim$ £26m, cover $\sim$ 1.2x, FCF after interest $\sim$ -£8m p.a., the RCF exhausts within a year — and the documents are cov-lite, so the first formal event may be the missed bullet itself. Probability-weighted, this is a watchlist credit *at origination*.

8.7 The decision — and how to write it

The Credit Memo sheet shows the full write-up: conclusion first, evidence after, five Cs, ranked ways out, downside case, triggers. The recommendation: **decline at the proposed structure**; re-engage only with (i) leverage reset to $\leq$ 4.0x lender-adjusted via equity or paydown, (ii) 5% amortisation plus a 50% cash sweep, (iii) maturity 2029+ with the RCF inside the term loan, (iv) quarterly maintenance covenants with addbacks capped at 10%, (v) a committed sponsor backstop for the refinancing. Note what the memo does *not* lead with: leverage. Leverage said "tight but market". The **cash flow statement said "cannot repay"; the debt schedule said "by September 2027"**. That ordering — cash first, schedule second, ratios third — is the transferable lesson of the entire case.

PART 9

Portfolio monitoring and early-warning indicators

Origination is a third of the job. Private credit positions cannot be sold on a screen; your exit from a deteriorating credit is early engagement, and early engagement depends on seeing deterioration in data before it becomes a default. Losses are usually not caused by the event that triggers default — they are caused by the four quarters of unwatched drift that preceded it.

9.1 The monitoring cadence

Frequency	Do
Monthly	Management accounts vs budget: revenue, EBITDA, cash, RCF utilisation, liquidity runway. 15-minute pass per name; flag any threshold trip (9.2).
Quarterly	Covenant certificates — recompute, never just file; full ratio refresh (the Part 3 five-cluster pass); headroom trend; red-flag checklist delta (Part 5); update the FCF walk if performance diverged from plan.
Annually	Audited accounts reconciliation vs management accounts; full re-underwrite: refreshed downside case, refinancing runway check (4.4), sector review, internal rating re-affirmation and IFRS 9 staging review.
Event-driven	CFO/auditor change, covenant amendment request, missed information deadline, sponsor news, sector shock: immediate file review, no waiting for the quarter.

9.2 Early-warning indicators — set numeric tripwires per name

- **Financial:** EBITDA miss >5–10% vs budget; covenant headroom <15%; cash conversion <65%; DSO +3–5 days QoQ; RCF utilisation above a name-specific line; liquidity below X months of fixed charges; any new "exceptional" item.
- **Behavioural:** reporting late by >5 days; unexplained restatement of prior management accounts; CFO/FD departure; auditor change; amendment or waiver request of any kind; sponsor going quiet.
- **Structural clock: 18 months before any maturity with no refinancing process demonstrably underway → automatic watchlist**, regardless of performance. This single mechanical rule catches the CRO's example deal even if every other indicator is missed.

9.3 Watchlist discipline and staging

A tripped EWI moves the name to the watchlist with a written action plan and an owner — the watchlist is a to-do list, not a naughty step. Map to impairment staging honestly: significant increase in credit risk (IFRS 9 Stage 2) is a *relative* deterioration test — a name can be Stage 2 while fully performing. Marks and internal ratings that lag known deterioration are how portfolio problems compound; the credit function's credibility with the CRO rests on ratings that move *before* payments do. When a name deteriorates: engage early (information first, then structure — fees/margin for waivers, covenant resets in exchange for amortisation/security, sponsor cash as the price of flexibility), and prepare the downside in parallel (security review, enforcement analysis, advisors identified) even while negotiating the upside.

PART 10

Checklists and the one-page crib sheet

10.1 The ten-question pre-screen (30 minutes, kills 30% of deals)

1. What, precisely, repays this loan — and is that source contractual, organic, or hoped-for?
2. Leverage on **statutory** EBITDA — and how big is the adjustment gap?
3. Interest cover at **forward** rates +200bps?
4. FCF after interest, last actual year — positive?
5. When is the maturity wall, and what is the organic repayment share by then?
6. Total liquidity today vs 12 months of obligations?
7. Three-year trend: gross margin, DSO, cash conversion — any two deteriorating?
8. What stands beneath me (real cash equity) and what is it worth in a downside?
9. Are there maintenance covenants, and is the EBITDA definition tight enough for them to mean anything?
10. Am I being paid for this risk relative to what the market pays for comparable risk?

10.2 The full underwriting checklist

Block	Items
Business & sector	Market position · concentration (customer/supplier) · cyclical & last-downturn performance · pricing power evidence · regulatory/ESG exposure
Earnings quality	Statutory vs adjusted vs covenant EBITDA · addback verification · QoE report read · accounting-policy review · audit opinion & notes
Ratio pass	Part 3 five clusters, 3–5yr trends, on all three EBITDAs, vs sector peers
Repayment	FCF walk to past maturity · organic repayment share · forward DSCR at the wall · refinancing underwrite · downside + reverse stress
Structure	Security perimeter & guarantor coverage · covenant package & headroom · EBITDA definition · leakage baskets · J.Crew/Serta blockers · sweeps & amortisation · intercreditor
Counterparties	Sponsor track record & dry powder · management quality & forecast credibility · auditors · (strategy-specific: LPs / GP / obligors / VC syndicate per Part 6)
Portfolio fit	Concentration vs limits (name/sector/vintage/strategy) · correlation with existing book · relative value
Decision hygiene	Pre-mortem written · conditions precedent listed · monitoring EWIs set with numbers · dissent recorded honestly

10.3 The one-page crib sheet

Ratio	Formula	Guide (senior corporate)	One-line meaning
Gross/Net debt ÷ EBITDA	on statutory EBITDA	<4x / 4–5.5x / >5.5x	Years of earnings owed
Adjusted gap	Adj ÷ Stat EBITDA – 1	<10% / 10–20% / >20%	How much of the credit case is spreadsheet
Interest cover	EBITDA ÷ cash interest	>3x / 2–3x / <2x	Affordability of the coupon
DSCR	(EBITDA–capex–tax) ÷ (interest+amort)	>1.2x / 1–1.2x / <1x	Can they pay the schedule
Forward DSCR at wall	same, maturity year, incl. bullet	explain any <1x	The CRO's-example catcher

FCF after interest	CFO – maint. capex – interest	>0 and ≥ amortisation	Cash actually available for principal
Cash conversion	CFO ÷ EBITDA	>85% / 60–85% / <60%	Is EBITDA real
DSO / DIO / DPO	AR, Inv, AP day-counts	trend ±5d = question	Where cash hides & earnings are made
Total liquidity	cash + undrawn committed	>10% revenue / 6m fixed charges	Survival in a bad quarter
LTV	net debt ÷ stressed EV	<50% / 50–70% / >70%	Cushion if it all goes wrong
Organic repayment share	cum. FCF ÷ principal due, over tenor	state it, always	How much is credit, how much is refi bet

AND THE SENTENCE TO KEEP ABOVE YOUR DESK

Leverage tells you a deal is tight. The **cash flow statement** tells you whether they can pay you back. The **debt schedule** tells you exactly when you will find out. Read them in that order, on every deal, every review.

PART 11

What "good credit quality" actually means — and how to grade it

11.1 The definition, from first principles

Credit quality is **the degree of certainty that a borrower will meet its obligations in full and on time across the range of plausible futures — combined with the quality of your recovery in the futures where it does not**. Every word is doing work. *Certainty*: credit quality is a probability distribution, not a snapshot — two borrowers with identical ratios today can have utterly different credit quality if one's cash flows are contracted and the other's are cyclical. *In full and on time*: a borrower that pays eventually, after an amend-and-extend, has already cost you — time is part of the promise. *Range of futures*: quality is measured in the downside, since the upside pays you nothing extra. *Recovery*: PD and LGD together, never one alone.

Two corollaries follow. First, **a good company is not the same as a good credit**. An outstanding business at 7.5x leverage with a wall in two years is a poor credit; a dull, slow-growth distributor at 2x with contracted revenues and amortising debt is an excellent one. Quality lives in the *relationship* between the business and its obligations, not in the business alone. Second, **credit quality is partly manufactured**. The same borrower is a different credit at different leverage, tenor, seniority and covenant packages — which is why underwriting (Parts 13–14) can create or destroy quality before a single trading day passes.

11.2 The five dimensions — what "good" looks like in the financials

Dimension	What it means	What "good" looks like in the data
Capacity	Cash generation comfortably exceeds debt service, with room for error.	FCF after interest consistently positive and $\geq$ scheduled amortisation; DSCR $>1.2x$; interest cover $>3x$ at forward rates; organic repayment share high or the refi underwritten.
Stability	The cash flows are predictable — low volatility, low cyclical, defensible position.	Revenue contracted/recurring or demonstrably defensive through the last downturn; margins stable within a narrow band over 5+ years; diversified customers; EBITDA trough-to-peak swing $<25\%$.
Flexibility	Levers exist when things go wrong: assets to sell, costs that flex, capex that can pause, owners with money.	Variable cost share high; capex has a genuine floor above collapse; separable assets; committed undrawn liquidity $>10\%$ of revenue; a sponsor with dry powder and reputational skin.
Structure	Your claim is positioned to detect trouble early and control the outcome.	Seniority and security over the value; maintenance covenants with 25–35% headroom; tight EBITDA definition; amortisation/sweeps forcing deleveraging; no wall inside the tenor.
Transparency	You can see the truth, quickly.	Clean audits, on-time monthly reporting, small management-to-audited gaps, forecasts that history shows were honest, management that volunteers bad news.

A credit can be investment-grade-like on four dimensions and be sunk by the fifth — Meridian's problem is overwhelmingly *capacity* (and the structure that failed to compensate for it). Grade each dimension separately before you blend them; blending first is how committees talk themselves into "on balance, fine".

11.3 Measuring it — ratings, PDs and models

The industry compresses credit quality into a **rating**: an ordinal grade mapped to a probability of default. You should be fluent in the mapping:

Grade (S&P scale)	Indicative 1-yr PD	Profile in ratio terms (stable mid-market corporate)
BBB– and above (investment grade)	< ~0.3%	Leverage <3x, cover >6x, positive FCF through the cycle, deep liquidity, undoubted refinancing access. Rare in private credit borrowers.
BB / BB–	~0.5–1%	Leverage ~3–4x, cover 3–5x, consistent FCF, modest walls. The best of a direct-lending book.
B+ / B	~2–4%	Leverage 4.5–6x, cover 2–3x, thin but positive FCF, refi-dependent bullets. The centre of gravity of sponsor-backed direct lending. Priced accordingly — your margin is mostly expected loss plus illiquidity.
B– / CCC+	~5–10%+	Cover <2x, FCF ≈ 0 or negative, addback-dependent ratios, liquidity eroding, wall inside 3 years. Meridian sits here — note that its <i>market label</i> at origination was "B+/4.2x adjusted".
CCC and below	>15%	Default is a scenario, not a tail: capital structure unsustainable absent a very good outcome.

- **Point-in-time vs through-the-cycle.** A PiT rating moves with conditions (useful for provisioning and early warning); a TTC rating abstracts from the cycle (useful for limits and appetite). Know which your internal scale is — a book that "upgrades itself" in a boom is applying PiT logic to TTC decisions, and will look over-rated exactly when the cycle turns.
- **Models are inputs, not verdicts.** Scorecards (weighted financial + qualitative factors), statistical PD models (Altman's Z-score is the grandfather: a weighted blend of working capital, retained earnings, EBIT, leverage and turnover ratios), and structural models (Merton: equity is a call option on the firm; default when asset value falls through the debt — the intuition behind "distance to default") each capture something. All of them miss what Parts 4–5 catch: maturity walls, addback games, liquidity cliffs. The analyst's judgment must be able to overrule the model — and every overrule must be documented, because override patterns are themselves an underwriting-quality metric (Part 13).
- **The output that matters is a sentence, not a number.** A complete quality assessment ends: *"This is a [grade] credit because [capacity evidence]; the rating is most vulnerable to [binding constraint]; it migrates down if [named, observable trigger]."* For Meridian: "B–/CCC+ credit: cover 1.7x and negative FCF against a 2027 wall; binding constraint is refinancing feasibility; migrates to CCC on any revenue miss >1% or DSO deterioration >3 days."

11.4 The in-depth assessment sequence

1. **Business risk grade:** industry cyclicality and structure, competitive position, diversification, size. (This caps everything: a weak business cannot be a strong credit at any but trivial leverage.)
2. **Financial risk grade:** the Part 3 ratio pass — capacity, leverage, liquidity — on lender-adjusted numbers, on trend, through the cycle.
3. **Anchor grade** = business × financial matrix (the agency method: a strong business carries more leverage at the same grade; a volatile one carries far less).
4. **Modifiers:** liquidity (can only drag down — weak liquidity caps the grade regardless of leverage), management/transparency, structure and documentation, sponsor/ownership.
5. **LGD separately:** seniority, security, collateral quality, jurisdiction → recovery band. Then $EL = PD \times LGD$ and the question that ends every assessment: **is the pricing paying for this EL plus the illiquidity plus a return on the risk?**

6. Trajectory and triggers: improving, stable or deteriorating — and the named events that would move the grade. A B+ improving is a different asset from a B+ deteriorating; most portfolio losses come from the second kind while the grade label looked identical.

PART 12

Stress testing any deal from the financials

12.1 What a stress test is for

A stress test is not a forecast and it is not decoration for the committee paper. It exists to answer four questions: **(1) what is the binding constraint** — the thing that breaks first (coupon? covenant? liquidity? the refinancing?); **(2) how much adversity survives** — the distance between today and that break; **(3) how fast** — the time from shock to impact, which sets how long you have to act; **(4) what is the cure** — and therefore what structure, pricing or decline the answer demands. If a stress test does not change the structure you ask for, the price you require, or the monitoring you set, it was theatre. The companion Excel now contains a live **Stress** sheet that implements everything in this part on the Meridian case.

12.2 The taxonomy

Type	What it does	When to use
Sensitivity	One variable at a time (revenue –10%; rates +200bps; DSO +5d), everything else held.	To locate <i>which</i> assumption carries the deal. Cheap; run half a dozen on every deal.
Scenario	A coherent story moving several variables together (recession: volumes down, margins squeezed, customers pay slower, refi markets shut).	The underwriting downside case. Variables must move <i>consistently</i> — a recession does not lower revenue while leaving DSO and refi spreads untouched.
Reverse stress	Start from failure (default, liquidity exhaustion, unfundable refi) and solve backwards for the smallest shock that causes it.	The discipline-keeper: expresses the deal's fragility as a single number ("this deal fails at a –X% revenue miss") that committees cannot wave away.
Portfolio stress	One macro scenario applied across the whole book; aggregates name-level results, exposes correlated exposures (same sector, same refi window, same LPs).	Second-line/CRO work: the sum of individually survivable deals can still be an unsurvivable book if their walls and sponsors coincide.

12.3 The seven-step method for any deal

- 1. Identify the value drivers.** From the financials: what actually moves cash? Volume, price/mix, input costs, wage inflation, FX, working-capital terms, capex intensity, base rates. Rank them by impact — a stress test that shocks the wrong driver is precision applied to irrelevance.
- 2. Calibrate the shocks from evidence, not vibes.** Three sources: (a) **the borrower's own history** — its worst year in the last ten, actual margin trough, actual DSO peak; (b) **sector history** — what did comparable companies do in 2008–09, 2020, 2022–23; (c) **macro mapping** — the borrower's revenue beta to GDP/consumer spending. House severity standards keep scenarios honest: a **mild downside ≈ a 1-in-7-year event** (a normal bad year), a **severe downside ≈ 1-in-25** (a proper recession plus one idiosyncratic blow, such as losing a top-3 customer). Every deal must survive mild intact and severe without becoming a workout.
- 3. Transmit through the P&L using operating leverage — the step juniors get wrong.** EBITDA does not fall 10% when revenue falls 10%. Split costs into variable and fixed:

$\Delta \text{EBITDA} \approx \Delta \text{Revenue} \times \text{contribution margin}$ (variable-cost basis) **$\pm \text{margin erosion} \times \text{remaining revenue}$**
 Meridian: contribution margin $\approx 29.5\%$, so a 15% revenue fall (–£38.7m) takes EBITDA down $\sim$ £11.4m — a **36% EBITDA fall from a 15% revenue fall**, before margin erosion. High fixed costs = high operating leverage = credit fragility that never appears in point-in-time ratios.

4. **Stress the balance sheet and liquidity, not just the P&L.** In a downturn: customers pay slower (DSO +3–7d absorbs cash), inventory sticks, suppliers tighten (DPO snaps back), the borrowing base shrinks exactly as the revolver is needed, and springing covenants can lock the undrawn RCF. Model the **liquidity runway month by month** — companies die of illiquidity on a Friday, not of leverage at year-end.
5. **Haircut management's counter-measures.** Management will claim cost cuts, capex deferral, working-capital release. Allow only actions that are specific, within their unilateral control, and evidenced by past delivery — and take 50% of those. Never allow the levers to offset the whole stress: a stress test where management actions neutralise the scenario is a base case wearing a costume.
6. **Read the outputs against the four questions.** For each scenario and year: interest cover, leverage, covenant headroom (or distance to the refi-market ceiling if cov-lite), FCF, liquidity runway in months, forward DSCR at the wall, and stressed enterprise value against your debt (LGD under stress — a 7x multiple in the base is 5.5–6x in the stress; sensitise both EBITDA *and* the multiple).
7. **Convert the answer into structure.** Severe case burns liquidity in ~8 months → minimum-liquidity covenant and a smaller RCF-to-wall gap. Refi ceiling breaks at a 1% revenue miss → amortisation and sweep to delever before the wall, or decline. The stress test's product is *terms*, not a chart.

12.4 Worked example — the Meridian Stress sheet

	Base FY26F	Mild (~1-in-7)	Severe (~1-in-25)
Assumptions	management plan	revenue –5%, GM –50bps, DSO +3d, funding +100bps, £1m credible actions	revenue –15%, GM –150bps, DSO +7d, funding +250bps, £2m actions
EBITDA (statutory basis)	35.1	31.1 (–11%)	22.4 (–36%)
Interest cover	1.85x	1.49x	0.94x — coupon unpayable
Gross leverage	5.4x	6.1x	8.4x
FCF after interest	~+1.0	–4.4	–17.1
Liquidity runway (£11.5m start)	n/a	~2.5 years	~8 months

And the reverse stress, which is the deal's verdict in three numbers: interest cover survives down to a **–21%** revenue shock; liquidity survives **~–18%** for a year; but the **5.5x refinancing ceiling is breached at roughly a –1% revenue miss**, because base leverage is already 5.4x statutory. The binding constraint is not the coupon and not even liquidity — it is the 2027 wall, and the margin for error against it is approximately zero. That sentence, produced mechanically by the method above, is what was missing from the file of the deal your CRO described.

12.5 Strategy-specific stress parameters

Strategy	What to shock (mild → severe)	The binding constraint to check
Direct lending	Revenue –5→–15%; margin –100→–300bps; DSO +3→+7d; rates/refi +100→+250bps; exit multiple –1→–2 turns.	Usually the refi wall or a covenant; check liquidity runway monthly through the trough.
Sub-lines / RCF	Largest 1–3 LPs excluded or defaulting; NAV –20→–40% (slows future calls); investment-period end pulled forward; GP key-person event.	Coverage ratio (unfunded/drawn) after exclusions; enforceability of calls under a stressed GP.
NAV financing	Portfolio marks –20→–40%; exit timings +12→+24 months; largest asset written to zero; asset-level lenders tightening.	LTV trigger cascade — do sweeps bite while value remains? Concentration after the good assets exit first.

Trade finance	Largest obligor defaults; dilution 2→3× historical; commodity price -20→-40% mid-transaction; tenor slippage +30→+60 days; a fraud event in the book.	Advance-rate cushion vs stressed dilution; insurance actually paying; self-liquidation genuinely occurring.
Venture debt	Next round delayed 6→12 months; burn +20→+40%; ARR growth halved; NRR -10pts; last-round valuation -30→-50%.	Runway vs your amortisation start date; investor-syndicate willingness under the down-round scenario.

12.6 The classic stress-testing failures

- **Anchoring:** "base minus a bit" — stresses calibrated to be survivable rather than plausible. Fix: calibrate from history (step 2), and let reverse stress expose the anchor.
- **Uniform mildness:** shocking ten variables by 2% each instead of the two that matter by a real amount. Fix: driver ranking (step 1).
- **Linear EBITDA:** ignoring operating leverage; assuming EBITDA falls pro-rata with revenue. Fix: contribution-margin transmission (step 3).
- **P&L-only stress:** ignoring working capital, borrowing-base shrinkage and covenant-triggered liquidity locks. Fix: step 4.
- **Heroic management:** letting cost actions offset the scenario. Fix: the 50% haircut rule (step 5).
- **Stress without consequence:** a red table in the appendix and unchanged terms. Fix: step 7 — the test's output is structure, price or decline.
- **One-year horizon on a five-year deal:** the wall is in year 3; stress to one year past final maturity, always.

PART 13

Assessing underwriting quality — the deal, the file, the book

13.1 What underwriting quality is

Underwriting quality is the quality of the **decision at the moment it was made** — with the information available then, was the analysis honest, the downside tested, the structure matched to the weaknesses found, the price adequate to the risk, and the decision taken inside a process with real challenge? It is explicitly *not* the same as outcome quality: good underwriting sometimes loses money (a 2% PD means one in fifty defaults even when you were right) and bad underwriting often gets paid for years before it doesn't. Judging underwriters by outcomes alone teaches them to be lucky, not good — but outcomes in aggregate are how you *calibrate* whether the process is honest. You therefore assess underwriting quality at two levels: the file (ex ante) and the book (ex post).

13.2 File-level review — the re-underwriting method

Take a sample of approved files (include some declines — they show you where the bar actually sits) and re-perform the analysis blind before reading the memo's conclusion. Then score the file against this checklist:

Test	What "pass" looks like
Independent numbers	EBITDA rebuilt from statutory accounts with addbacks itemised, evidenced and haircut — not the sponsor's figure pasted in. Working capital normalised. The three EBITDAs reconciled.
Repayment named and ranked	The ways out stated in order with the organic repayment share computed; if refi-dependent, the refinancing underwritten (Part 4.4), not assumed.
Downside genuinely run	A calibrated scenario (12.3), with the binding constraint and reverse-stress number identified — not "sensitivity: EBITDA –10%" as a table nobody discusses.
Structure answers the analysis	Each identified weakness maps to a term: weak FCF → amortisation/sweep; addback risk → tight definition and caps; liquidity risk → minimum-liquidity covenant. A memo that lists risks and a structure that ignores them is decoration.
Pricing tied to risk	Spread vs internal EL + capital + illiquidity, and vs comparable market deals — with the comparison shown.
The risks section is real	Specific, quantified, uncomfortable — not boilerplate ("execution risk exists"). The best single tell of file quality: could the risks section have been written without reading the financials? If yes, it was.
Challenge visible	Committee questions and answers minuted; conditions imposed and <i>verifiably met before funding</i> ; any dissent recorded. Overrides of model ratings or policy exceptions documented with reasons.
Would the file catch the CRO's example?	The acid test: does the file contain the two-line lender's arithmetic (Part 1.3) and a forward DSCR at the wall? If a file can be approved without those, the failure mode is open.

13.3 Book-level metrics — reading underwriting quality out of portfolio data

Metric	How to compute	What it tells you
Vintage loss vs priced EL	Realised + expected losses by origination year vs the EL implied at underwriting, tracked as vintages season.	The ultimate calibration: losses persistently above priced EL = systematic optimism; far below = possibly over-declining good risk (also a cost).

Rating migration by vintage	% of each vintage downgraded ≥ 1 notch within 12/24 months of close.	Deals that deteriorate immediately after close were mis-graded at close — the fastest honest signal, available years before losses.
Early amendments	% of deals seeking covenant reset/waiver/amendment within 18 months.	An amendment at month 12 is an underwriting error that was negotiated instead of recognised.
Addback realisation	For each deal: covenant-EBITDA addbacks at close vs statutory EBITDA actually achieved 1–2 years on.	Measures how much of the book's "EBITDA" was ever real. Portfolio-level addback realisation below ~50% means headline leverage understates the book by a turn.
Forecast accuracy scoring	Underwriting base case vs actuals (revenue, EBITDA, leverage) at year 1/2, by deal team.	Separates optimistic teams from honest ones; feeds back into how much weight committees give each team's base case.
Exceptions & overrides	Rate of policy exceptions and model-rating overrides; subsequent performance of excepted/overridden deals vs the rest.	Exceptions that outperform = policy too tight. Exceptions that underperform = the exception process is a leak. Either way, you learn.
Time-to-watchlist	Lag between first objective EWI trip and formal watchlist entry.	Long lags mean monitoring sees problems and does not act — an underwriting culture problem, not a data problem.
Decline rate & shape	% of screened deals declined, and at what stage.	A near-100% approval rate at committee means the real decisions happen earlier, unminuted — governance is happening somewhere you cannot see it.

13.4 Behavioural red flags of weak underwriting

- **Volume incentives:** originators paid on deployment with no deferral against performance — the single strongest predictor of vintage trouble, visible years in advance.
- **Copy-paste memos:** identical risk sections across deals; the analysis is a template, and templates do not read cash flow statements.
- **Base-case-only culture:** downside cases missing, or present but never referenced in the recommendation.
- **Structure creep between approval and close:** terms approved by committee quietly diluted in documentation ("market flex", "final docs"); compare signed docs to the approved term sheet on a sample.
- **Conditions waived at close:** CPs imposed by committee then waived by the deal team under completion pressure.
- **"Market standard" as an argument:** the market's average discipline is procyclical by construction; it is loosest exactly at the top.
- **Serial re-approvals:** the same credit returning for increases with the same unresolved weaknesses, approved each time because "we're already in".

PART 14

Assessing an underwriting framework — the CRO's lens

14.1 The framework is a system, and it is only as good as its feedback loop

Individual underwriting quality (Part 13) asks "was this decision good?". A framework assessment asks "does the *system* reliably produce good decisions — including when volumes are high, markets are hot, and the people are new?" A framework has seven components in a loop; assess each, then assess the loop itself:

Component	What good looks like	How to test it
1. Risk appetite	Written, measurable, linked to capital and funding: leverage caps by strategy, single-name/sector/vintage concentration limits, minimum pricing floors, cov-lite tolerance, wall concentration limits. Board-approved, reviewed annually.	Pick five recent deals: can you tell, from the appetite statement alone, whether each was inside it? If appetite requires interpretation per deal, it is a preference, not a limit.
2. Credit standards / policy	Minimum mandatory analysis per deal: statutory-based EBITDA rebuild, the FCF-vs-maturity arithmetic, a calibrated downside, forward DSCR at any wall, addback caps, valuation standards, ESG/sector exclusions. The Part 10 checklist, made compulsory.	Policy-vs-practice gap sampling: score 10 files against the written standard. The gap, not the policy document, is the framework.
3. Approval governance	Independent credit function with veto (not advisory voice); committee with quorum, recorded votes, minuted challenge; delegated-authority matrix by size/risk/exception; second-line present at the table, not reviewing after close.	Read six months of minutes: count real questions, conditions imposed, declines. A committee that has declined nothing has decided nothing. Check any deal approved against a credit-function objection — who overrode, and what happened?
4. Incentives	Origination pay deferred and clawback-linked to vintage performance; credit function's pay independent of deployment; no one marks their own homework (deal team ≠ rating assigner).	Follow the money: ask HR for the actual bonus mechanics, not the policy's description of them.
5. Data & models	Single source of borrower financials; rating models validated and back-tested on the firm's own outcomes; portfolio MI that can answer "what is our exposure to walls in 2027?" same-day.	Ask three "simple" portfolio questions and time the answers. If the wall-concentration question takes a week and a spreadsheet hunt, monitoring is structurally blind.
6. Monitoring & EWIs	The Part 9 machinery: numeric tripwires per name, watchlist with owners and action plans, the mechanical 18-months-to-maturity rule, staging that moves before payments do.	Take the last five defaults/workouts: when did each first trip an objective EWI, and when did the firm act? The gap is the framework's reaction time.
7. Feedback loop	Every workout produces a written lessons-learned that names the underwriting gap; Part 13's book metrics reported quarterly to the risk committee; policy updated on a cadence with a change log; forecast-accuracy scores fed back to teams.	Ask for the last three lessons-learned memos and the policy changes they produced. No memos, or memos without consequences = the system cannot learn; it will repeat its losses with new names.

14.2 The single best framework test: trace the deal that got through

Take your CRO's example — the borrower that could not earn its way to the repayment — and walk it through the framework asking, at each gate, "*would this gate have caught it, mechanically, without needing a hero?*"

- Standards gate:** did policy *require* the FCF-vs-maturity arithmetic and a forward DSCR at the wall? If not, the analysis gap is a policy gap — fix the standard, not the analyst.
- Challenge gate:** the memo presumably showed the bullet and the thin FCF somewhere. Did committee ask the repayment question? If the information was present but unquestioned, the failure is governance,

not analysis.

3. **Structure gate:** did policy require amortisation/sweeps or a sponsor backstop where organic repayment was below a threshold? A framework that lets pure refi bets close as bullets with no compensating structure has priced hope.
4. **Monitoring gate:** even after approval, the 18-month maturity rule would have forced the name onto the watchlist with time to act. Did it exist? Was it triggered? Was it actioned?

A sound framework has **redundancy**: the deal must pass four independent gates, so one tired analyst or one deferential committee does not become a loss. When you present to your CRO, this trace — "here are the four places our framework now catches that deal" — is the most convincing artefact you can bring.

14.3 A maturity model for the framework

Component	Weak	Developing	Strong
Appetite	Aspirational prose	Limits exist, exceptions frequent and untracked	Measurable, capital-linked, exceptions capped/reported/priced
Standards	Analyst discretion	Checklist exists, sampling shows gaps	Mandatory content enforced; downside & wall arithmetic compulsory
Governance	Originators approve own deals	Committee exists, challenge thin, declines rare	Independent veto, minuted challenge, decline rate healthy
Incentives	Paid on volume	Deferral exists, no clawback link to vintages	Deferred, vintage-linked, credit pay independent
Data/models	Spreadsheets per team	Central data, models unvalidated	Validated models, same-day portfolio queries, override tracking
Monitoring	Annual review only	Quarterly ratios, subjective watchlisting	Numeric EWIs, mechanical maturity rule, staging moves early
Feedback	Losses blamed on markets	Post-mortems occur, change nothing	Lessons-learned → policy change log; book metrics to risk committee quarterly

14.4 Ten questions a board-level CRO asks (keep these in your pocket)

1. What is our organic repayment share, book-wide — how much of this portfolio is credit, and how much is a refinancing bet?
2. Show me the maturity-wall profile of the whole book by year. Which vintages and sectors coincide?
3. What fraction of covenant EBITDA across the book is addbacks — and what is our realised addback-realisation rate?
4. Which names would breach a minimum-liquidity test in our severe scenario, and what is their aggregate exposure?
5. What was our rating-migration rate within 24 months of close, by vintage and by deal team?
6. How many deals did we decline last quarter, and why? Show me one we declined that a competitor closed.
7. Which approvals overrode a credit-function objection in the last two years, and how are they performing?
8. When a name trips an EWI, what is our median time to watchlist and to action?
9. What did our last workout teach us, and which sentence of policy changed because of it?
10. If the deal that started this conversation were submitted today — which gate stops it?

THE SUMMARY OF PARTS 11–14

Credit quality is certainty of full, timely repayment across plausible futures plus recovery where it fails — graded on capacity, stability, flexibility, structure and transparency, and expressed as a rating with a named binding constraint. **Stress testing** is how you locate that constraint: calibrated shocks, transmitted through operating leverage and the balance sheet, haircut for heroism, converted into terms. **Underwriting quality** is decision quality ex ante, audited by re-performing files and by reading the book's vintages, migrations, amendments and addback realisation. An **underwriting framework** is the system that makes good decisions the default — appetite, standards, governance, incentives, data, monitoring, feedback — and it is proven by one test: the deal that once got through can no longer pass any gate silently.

GLOSSARY

Terms used in this handbook

Term	Definition
Addback	Cost added back to EBITDA as supposedly one-off or non-cash; the primary channel of EBITDA inflation in leveraged credit.
Advance rate	% of eligible collateral value a lender will fund (borrowing-base facilities, fund finance, trade finance).
Amortisation	Scheduled contractual repayment of principal over the loan's life (vs a bullet).
ARR / NRR / GRR	Annual recurring revenue; net revenue retention (incl. upsell); gross revenue retention (churn only) — venture-debt revenue-quality metrics.
Borrowing base	Formula-driven facility limit computed from eligible collateral less reserves.
Bullet	Principal repaid in one payment at maturity; converts repayment risk into refinancing risk.
Burn / runway	Net monthly cash consumption; months of cash remaining at current burn.
Capital call / sub-line	Fund's demand on LPs to fund commitments; facility secured on the right to make such calls.
Cash conversion cycle	DSO + DIO – DPO: days of the trading cycle needing finance.
Cov-lite	Loan with incurrence covenants only — no quarterly maintenance tests.
Dilution (trade finance)	Reduction of receivable value from returns, disputes, rebates, offsets.
DSCR / FCCR	Debt-service and fixed-charge cover ratios — cash earnings vs contractual payments.
EAD / PD / LGD / EL	Exposure at default; probability of default; loss given default; expected loss = $PD \times LGD \times EAD$.
EBITDA (statutory / adjusted / covenant)	Earnings before interest, tax, D&A — per audited accounts; per management addbacks; per loan-document definition.
Equity cure	Sponsor's contractual right to inject cash to remedy a covenant breach.
Excess-cash-flow sweep	Mandatory prepayment of a % of annual FCF; forces deleveraging before maturity.
Incurrence / maintenance covenant	Tested only on borrower action / tested every period regardless.
J.Crew / Serta	Shorthand for asset-stripping to unrestricted subsidiaries / majority-lender uptiering priming minorities.
LP / GP / LPA	Fund investor / fund manager / the fund's limited partnership agreement.
LTV	Loan (net debt) as % of asset or enterprise value.
Maturity wall	Concentration of contractual maturities in a short window.
NAV facility	Loan to a fund secured on portfolio net asset value; repaid from realisations.
OID / MFN / MAC	Original issue discount; most-favoured-nation pricing protection; material adverse change clause.
PIK	Payment-in-kind interest — accrues to principal instead of being paid in cash; compounds leverage.
QoE	Quality-of-earnings report: due-diligence rebuild of sustainable EBITDA and normalised working capital.
RCF	Revolving credit facility — committed line drawn/repaid at will; in fund finance, shorthand for the subscription line.
Springing covenant / maturity	Covenant or maturity that activates on a trigger (e.g. RCF utilisation %; failure to refinance another instrument by a date).
Unitranche	Single-tranche loan blending senior and junior risk/pricing; standard direct-lending format.
Uncalled / unfunded commitment	LP capital committed but not yet drawn — the collateral of a subscription line.
Watchlist / Stage 2	Enhanced-monitoring status; IFRS 9 classification for significantly increased credit risk while still performing.

Ways out	Ranked repayment sources: cash flow, refinancing, support, collateral.
Addback realisation	Share of underwriting-time EBITDA addbacks that later appear in actual statutory results; a book-level underwriting-quality metric.
Binding constraint	The element of a credit that breaks first under stress (coupon, covenant, liquidity, or refinancing ceiling); the primary output of a stress test.
Contribution margin	Revenue minus variable costs, as % of revenue; determines how revenue shocks transmit to EBITDA (operating leverage).
Distance to default	Structural-model (Merton) measure of how far asset value sits above the default point, in volatility units.
Outcome bias	Judging decision quality by results rather than by the information and process at the time — the central error in assessing underwriters.
PiT / TTC rating	Point-in-time rating (moves with conditions) vs through-the-cycle rating (abstracts from the cycle).
Reverse stress test	Solving backwards from failure for the smallest shock that causes it; expresses fragility as one number.
Scorecard	Weighted expert model combining financial ratios and qualitative factors into a rating.
Vintage analysis	Tracking loans grouped by origination year as they season; isolates underwriting quality from the cycle.